signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
t_max = 5; // End time
dt = 0.01; // Time step
t = t_min:dt:t_max; // Time vector
// Define the impulse function
x_impulse = zeros(1, length(t));
x_impulse(t == 0) = 1; // Approximate impulse function
// Initialize the output signal with zeros
y_impulse = zeros(1, length(t));
// Define the system parameters
a = 2; // Coefficient for y(t)
b = 1; // Coefficient for x(t)
// Compute the impulse response using the Laplace Transform approach
// Analytical solution: y(t) = (b/a) * exp(-a*t) for t >= 0
for i = 1:length(t)
if t(i) >= 0
y_impulse(i) = (b / a) * exp(-a * t(i));
end
end
// Plot the input and output signals for impulse response
clf; // Clear the current figure
subplot(2,1,1);
plot(t, x_impulse, 'b');
xlabel('t (Time)');
ylabel('x(t)');
title('Input Signal x(t) - Impulse Function');
subplot(2,1,2);
plot(t, y_impulse, 'r');
xlabel('t (Time)');
ylabel('y(t)');
title('Impulse Response y(t) of the System');

```

